PhD Fellowship Proposal: Ethereum Foundation 2026

Project Abstract

This research addresses the high transaction costs and operational inefficiencies faced by MSMEs in Nairobi, Kenya, by exploring Ethereum-based P2P payment systems as a decentralized alternative to centralized mobile money. While Kenya leads in digital finance, small enterprises remain burdened by high fees and a lack of cross-border interoperability. This study empirically investigates the determinants of adopting Ethereum L2/stablecoin solutions and evaluates their impact on financial inclusion, liquidity management, and credit accessibility for 460 MSMEs.

Objectives

- **Identify Barriers:** Determine the specific technological, social, and regulatory factors that prevent or encourage the adoption of Ethereum-based payments among Kenyan MSMEs.
- **Measure Impact:** Quantify the shift in "financial inclusion" (defined as reduced transaction overhead and improved access to digital credit) through the use of programmable settlement.
- **Inform Policy:** Provide evidence-based recommendations to the Central Bank of Kenya (CBK) on integrating Ethereum-based primitives into the national digital economy strategy.

Outcomes

This project benefits the Ethereum ecosystem by providing a **large-scale, real-world pilot study** on how stablecoins and L2s perform in one of the world's most active digital payment markets. It bridges the gap between Ethereum's technical capabilities and the "Software-Defined Operations" needed by businesses in emerging markets, driving long-term involvement and adoption in Sub-Saharan Africa.

Grant Scope

The research focuses on the "Application Layer" of Ethereum. I will research the adoption cycle of stablecoins (USDC/USDT) and smart contract escrows for everyday business processes.

- **Expected Output:** A formal academic paper (targeted for *MIS Quarterly* or *Journal of Fintech*), an open-access dataset of anonymized MSME payment behaviors, and a set of deployable smart contract primitives for SME escrow and dispute resolution.

Related Work

- **Existing Research:** My work builds on the **UTAUT (Unified Theory of Acceptance and Use of Technology)** framework by Venkatesh (2020) and research on DeFi for financial participation (Mhlanga, 2023).
- **The Gap:** Most existing research focuses on crypto-speculation or supply chain logistics. There is a critical lack of empirical data on **everyday P2P business transactions** using Ethereum in high-density MSME hubs like Nairobi.

About the Applicant

I am a PhD candidate in Business Administration and Management at Mount Kenya University, holding an MBA in Strategic Management and a graduate of Computer Science. I am a software developer with hands-on experience in blockchain development (e.g., the "FreelanceFlow" project, which utilizes USDC escrows). Being based in Nairobi allows me direct access to the target population of 460 MSMEs.

- **Relevant Links:** https://github.com/TarusNicky8/PhD-Ethereum-MSME-Nairobi , Nicodemus Kiptoo (@nicodemuskipto0) / X
- **Supervisor:** Dr. Charles Karuga, Director of Nairobi Campus, Mount Kenya University. Contact: CKaruga@mku.ac.ke

Methodology

The research follows a **positivist paradigm using a quantitative approach**.

1. **Survey Instrument:** A structured Likert-scale questionnaire validated for the Nairobi context.
2. **Sampling:** Stratified random sampling of 460 MSMEs across Trade, Service, and Manufacturing sectors.
3. **Statistical Modeling:** Data analysis using **Partial Least Squares Structural Equation Modeling (PLS-SEM)** via SmartPLS 4 to determine the weight of each adoption factor.

Timeline

Milestone	Title	Summary of Work
Milestone 1	Literature & Instrument Validation	Refining the survey based on Ethereum L2 specifics and local pilot tests. (approx. 150 hours)
Milestone 2	Fieldwork & Data Collection	Direct engagement and data gathering from 460 MSMEs in Nairobi. (approx. 400 hours)
Milestone 3	Data Analysis & PLS-SEM	Running the structural models to find determinants and impact. (approx. 200 hours)
Milestone 4	Final Paper & Primitive Design	Finalizing the academic manuscript and open-sourcing the SME escrow primitives. (approx. 250 hours)

Budget: The $24,000 stipend will be allocated toward field researcher logistics, data analysis software licenses, and academic publication fees, distributed over the 1,000 estimated research hours ($24/hour).